

Mathematical Analysis of Tic – tac – toe Grid

Abstract :

This paper presents a mathematical analysis of the Tic-Tac-Toe game on an $N \times N$ grid. The total number of possible winning conditions is derived using a simple counting approach. In addition, each winning line—rows, columns, and diagonals—is represented using vector parametric equations.

This provides a structured way to understand the geometry of the grid and the formation of winning patterns. The approach connects combinatorics and coordinate geometry to explain the game mathematically.

Concepts Used :

Co-ordinate Geometry, Vector and Parametric Equation, Grid System

Introduction :

Tic – tac – toe is a simple and widely known game played on a grid. While it is usually studied as a strategy game problem, it also has a clear mathematical structure.

In this paper, the game is analyzed using mathematics. The total number of winning conditions is derived, and each winning lines is represented using coordinate geometry. By expressing the rows, columns and diagonals as parametric equations, the structure of the grid becomes more clear and systematic.

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1) Vector(Parametric) Equations of each winning lines :

For Diagonal(L_1) Eqs,

L_1 is passing through $(0, 0)$ and $(N-1, N-1)$,

Vector Parametric Eq of L_1 ,

$$\vec{d} = \frac{N-1}{N-1} = (1, 1)$$

$$\begin{aligned}\vec{r}(t) &= \vec{a} + \vec{d}t \\ &= (0, 0) + (1, 1)t\end{aligned}$$

$$\therefore \vec{r}(t) = (t, t)$$

$$\therefore X_t = t, Y_t = t \quad \text{..... eq(i)}$$

For Diagonal(L_2) Eqs,

L_2 is passing through $(N-1, 0)$ and $(0, N-1)$,

Vector Parametric Eq of L_2 ,

$$\vec{d} = \frac{-(N-1)}{N-1} = (-1, 1)$$

$$\begin{aligned}\vec{r}(t) &= \vec{a} + \vec{d}t \\ &= (N-1, 0) + (-1, 1)t\end{aligned}$$

$$\therefore \vec{r}(t) = (N-1-t, t)$$

$$\therefore X_t = N - (t+1), Y_t = t \quad \text{..... eq(ii)}$$

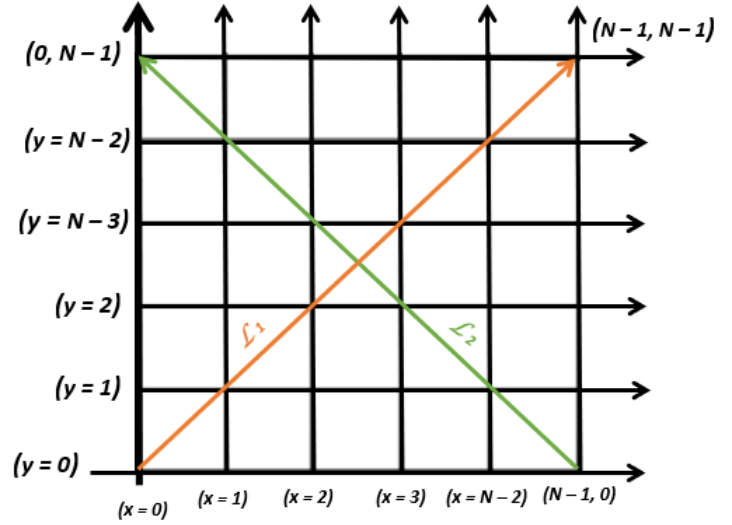


Figure 1: Tic-tac-toe grid in Graphs.

Similarly,

For each Row Lines,

$$(X_t, Y_t) = (t, k)$$

$$\therefore X_t = t, Y_t = k \quad \text{..... eq(iii)}$$

Also,

For each Column Lines,

$$(X_t, Y_t) = (k, t)$$

$$\therefore X_t = k, Y_t = t \quad \text{..... eq(iv)}$$

2) Total no of winning Combinations :

$$\text{No of Columns (C)} = N$$

$$\text{No of Diagonals (D)} = 2$$

$$\text{No of Rows (R)} = N$$

$$\begin{aligned}\text{Total winning Combinations} &= \text{Rows} + \text{Columns} + \text{Diagonals} \\ &= N + N + 2\end{aligned}$$

$$\therefore \text{Total no of winning Combinations} = 2(N+1) \quad \text{..... eq(v)}$$

Here,

$$\text{Tic - tac - toe (N)} = 5$$

For Diagonal Co-ordinates,

$$\vec{r}(t) = (t, t)$$

$$\text{Range of } t : (0 < t < 5)$$

Then,

$$\text{When } t = 0, \vec{r}(0) = (0, 0)$$

$$\text{When } t = 1, \vec{r}(1) = (1, 1)$$

$$\text{When } t = 2, \vec{r}(2) = (2, 2)$$

$$\text{When } t = 3, \vec{r}(3) = (3, 3)$$

$$\text{When } t = 4, \vec{r}(4) = (4, 4)$$

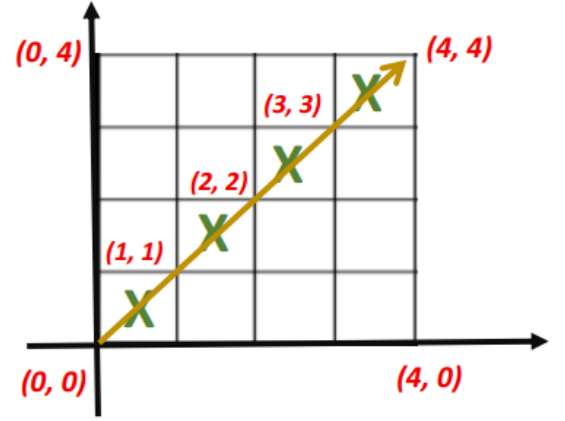


Figure 1: Diagonal Coordinates in Grid

$$\therefore \text{Co-ordinates of Diagonal} = (0,0), (1,1), (2,2), (3,3), (4,4)$$

For Anti - Diagonal Co-ordinates,

$$\vec{r}(t) = \{(N - (t + 1), t)\}$$

$$\text{Range of } t : (0 < t < 5)$$

Then,

$$\text{When } t = 0, \vec{r}(0) = (4, 0)$$

$$\text{When } t = 1, \vec{r}(1) = (3, 1)$$

$$\text{When } t = 2, \vec{r}(2) = (2, 2)$$

$$\text{When } t = 3, \vec{r}(3) = (1, 3)$$

$$\text{When } t = 4, \vec{r}(4) = (0, 4)$$

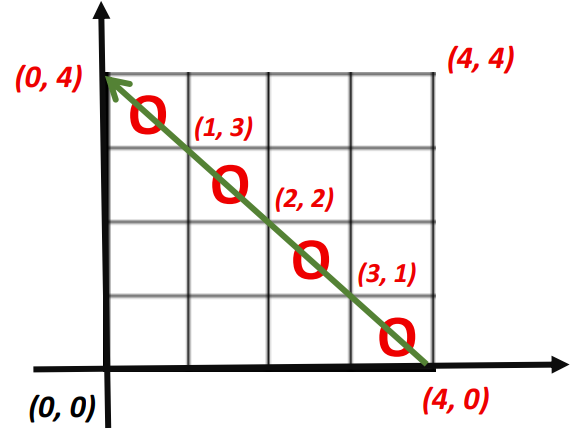


Figure 2: Anti - Diagonal Coordinates

$$\therefore \text{Anti - Diagonal Coordinates} = (4,0), (3,1), (2,2), (1,3), (0,4)$$

For total no of wining Conditions,

We Know,

$$\text{Total Winning Conditions} = 2(N + 1)$$

$$= 2(5 + 1)$$

$$= 12$$

$$\therefore \text{Total winning conditions} = 12$$

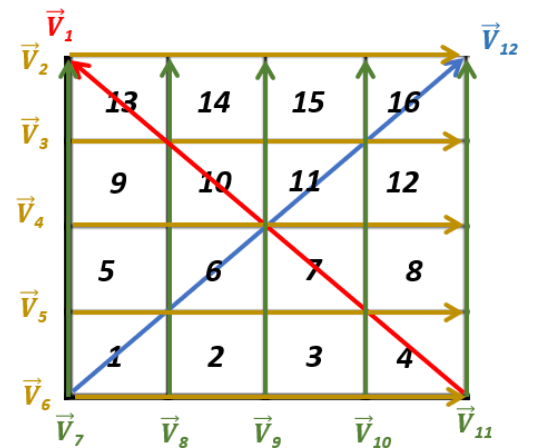


Figure 3: All Winning Vector St Lines